

Project code: D3

Project title: KAGGLE Customer Shopping Behaviour

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Link to the repository: <https://github.com/MirruMal/CustomerShoppingBehaviour>

Business understanding

1. Identifying Our Business Goals

Background

This project is based on a KAGGLE Customer Shopping Behaviour dataset from the United States of America. It contains variables such as age, gender, item purchased, category, purchase amount, location, size, color, season, review rating, subscription status, shipping type, discount appliance, promo code usage, previous purchases, payment method and frequency of purchases. The dataset reflects typical e-commerce or retail shopping patterns. We perform this analysis from the viewpoint of a retail business that wants to better understand its customer base and improve decision-making using data.

Business Goals

The business aims to understand what drives customer shopping decisions in order to improve personalization, optimize marketing, and better serve customer needs. Specifically, the goals are:

1. Identify meaningful customer segments based on behaviour (for example: high spenders, seasonal shoppers, discount-driven customers).
2. Understand which factors most strongly influence purchase amounts.
3. Predict which customers are likely to return (customer loyalty).

Business Success Criteria

The project will be considered successful if:

- We produce clear and interpretable customer segments that marketing teams can use in real decision-making.

- We identify at least 2–3 strong patterns that explain higher spending and can guide pricing or product strategies.
- We create a simple but functional prediction model that determines customer loyalty with reasonable accuracy (perhaps above 70%).
- Insights are actionable, not purely descriptive.

2. Assessing Our Situation

Inventory of Resources

- **Dataset:** Customer Shopping Behaviour dataset from Kaggle:
<https://www.kaggle.com/datasets/rehan497/customer-shopping-behavior-dataset>
- **Tools:** Python, Pandas, scikit-learn, Matplotlib/Seaborn, Jupyter Notebook.
- **Team skills:** Basic experience with data cleaning, visualization, clustering, and simple machine learning models.
- **Time:** Limited timeframe suitable for simple models and visualizations. It will probably take us 14 hours or more to complete this project.

Requirements, Assumptions, and Constraints

- The dataset must be cleaned before any analysis can be conducted.
- We assume the dataset is representative enough to identify general behavioural patterns.
- No private or sensitive information is included, which allows unrestricted analysis.
- Constraints: limited data size, simplified features, and no access to real-time business systems.

Risks and Contingencies

- **Risk:** Clustering may not produce clearly separable segments.
Contingency: Create manually defined personas instead of relying on algorithmic clusters.
- **Risk:** Spending patterns may show weak correlations.
Contingency: Use multiple visualizations and explore additional combinations of variables.
- **Risk:** Loyalty prediction model may underperform.
Contingency: Simplify the target definition or try a different model (maybe the logistic regression).

Terminology

- **Customer Segment:** A group of customers with similar behaviours or patterns.
- **Loyalty:** The likelihood that a customer continues buying, defined by subscriptions or high previous purchases.
- **K-Means:** A clustering algorithm for grouping data points.
- **Purchase Amount:** Total value of a customer's transaction.

Costs and Benefits

Costs:

- Time spent cleaning and interpreting data.
- Technical resources (software, analysis time).

Benefits:

- Better understanding of customer behaviour.
- Insights that help businesses reduce irrelevant marketing and focus on real customer needs.
- Foundations for more advanced customer analytics in a real company context.

3. Defining Our Data-Mining Goals

Data-Mining Goals

The analytical goals linked directly to the business goals are:

1. **Customer Segmentation:** Use K-Means or manual personas to categorize customers based on numerical and categorical behavior variables.
2. **Factor Analysis for Spending:** Identify which features correlate most strongly with higher purchase amounts using grouped visualizations and correlation analysis.
3. **Loyalty Prediction:** Train a simple supervised learning model (Decision Tree) to predict whether a customer is likely to return.

Data-Mining Success Criteria

- The segmentation method must produce groups that are distinct enough to be interpreted and described clearly.
- Spending analysis must reveal at least a few meaningful and explainable patterns.
- The loyalty model should demonstrate reasonable predictive performance (like above-baseline accuracy) and provide feature importance insights.
- All results must be understandable to a non-technical audience.

Data understanding

Outlining Data Requirements

To support the project goals, customer segmentation, understanding behavioral influencers of purchase amounts, and predicting loyalty, we require demographic attributes (such as age and gender), transactional information (such as items purchased and purchase amounts), engagement indicators (review ratings, subscription status), and behavioral context (season, shipping type, discounts, and payment method). We also need a variable that reflects customer history, such as the number of previous purchases, to help analyze loyalty patterns. The dataset must include enough cases and coverage to represent different customer types and shopping situations.

Verifying Data Availability

The dataset contains thousands of purchase records from customers across the United States. Each record represents a single transaction and includes the full set of attributes needed for segmentation, exploratory analysis, and predictive modeling. The dataset loads successfully into our analysis environment, and all fields appear to be accessible and compatible with standard data-mining tools. No structural issues or unreadable formats were encountered on initial observation.

Defining Selection Criteria

Since the project aims to analyze general customer behavior, the full dataset is included at this stage. All rows correspond to valid customer transactions, and no specific filtering criteria were applied initially. All fields are retained for exploration, even if some may later prove irrelevant for modeling. Fields such as Color or Size may not contribute to predictive goals, but they remain included until a later selection step. The dataset covers a wide variety of products and customer types, making it suitable for the intended analyses.

Describing Data

The dataset consists of 17 columns, representing demographic, behavioral, contextual, and transactional information. A brief description of the key fields:

- Customer ID: Unique identifier for each customer.
- Age, Gender, Location: Demographic information relevant for segmentation.
- Item Purchased, Category, Size, Color: Product attributes.
- Purchase Amount (USD): Numeric field representing spending per transaction.
- Season: The time of year of the purchase.
- Review Rating: A numerical satisfaction rating.
- Subscription Status: Indicates loyalty-program enrollment.
- Shipping Type: Delivery choice (for example: Express, Free Shipping).
- Discount Applied, Promo Code Used: Indicators of promotional engagement.

- Previous Purchases: Total number of past purchases by the customer.
- Payment Method, Frequency of Purchases: Behavioral descriptors.

Exploring Data

Initial exploration reveals several important characteristics of the dataset:

- Demographics: Ages range widely across adults, which supports generalizable segmentation. Gender distribution appears balanced. Locations are spread across various U.S. states.
- Purchase Amount: Values vary from low-cost purchases to high-value items, allowing analysis of spending patterns.
- Customer History: Previous Purchases range from first-time buyers to highly loyal customers with dozens of past transactions, providing useful variation for predicting loyalty.
- Promotional behavior: Discount usage and promo code usage are frequent, suggesting many customers are sensitive to promotions.
- Product characteristics: Multiple product categories and item types are represented, indicating diverse shopping patterns.
- Engagement: Review ratings typically fall in the 2–5 range, suggesting moderate satisfaction levels; subscription status is common among customers.

Verifying Data Quality

Preliminary quality checks show that the dataset is generally well-structured, though there still might be several issues that require our attention, which we'll have to check before doing anything important with the data:

- Certain categorical fields may include variations in spelling or formatting and may need standardization (such as Color: Gray or Grey).
- Purchase Amount and Previous Purchases could contain extreme values that must be examined to determine whether they represent legitimate customer behavior or data errors.
- Some variables, while useful for descriptive analysis, may not contribute to predictive tasks and may be excluded later.

Planning the project

List of Tasks				
Task	Methods and Tools	 Person responsible	 State	Time
1. Data Understanding and Cleaning Explore the dataset, understand each column, handle missing values, convert data types, and prepare clean data for analysis. - This task is crucial; errors here will affect the rest of the project. - Make sure columns like Purchase Amount, Age, Frequency are numeric.	Pandas Jupyter Notebook Data inspection (describe, info, unique values) Missing value handling Basic visualizations (histograms)	 Inim...	Pooleli ▾	Mirelle: 3 hours Anne-Liine : 3 hours
2. Customer Segmentation Group customers based on shopping behavior patterns such as Frequency, Previous Purchases, Subscription Status, Season, and Purchase Amount K-Means requires numeric data, so categorical variables need encoding. - If segmentation becomes too complex, fallback to manually defined personas.	K-Means clustering Alternatively: manual creation of 3–4 customer personas StandardScaler (if needed) Visualization with bar charts	 Inim...	Ei ol... ▾	Mirelle: 3 hours Anne-Liine : 3 hours

List of Tasks

Task	Methods and Tools	 Person responsible	 State	Time
3. Spending Behaviour Analysis Analyze which factors influence higher purchase amounts (age, gender, category, subscription status, previous purchases). - Focus on clear comparisons (for example, category vs. purchase amount). - Identify at least 2–3 strong relationships.	Grouped bar charts Boxplots Correlation matrix Pandas groupby Matplotlib / Seaborn	 Inim...	 Ei ol... ▾	Mirelle: 3 hours Anne-Liine : 3 hours
4. Customer Loyalty Prediction Model Predict whether customers are likely to return. Define loyalty using subscription status and previous purchases. Train a simple model. Clearly define what “loyal” means <u>before training</u>.	Decision Tree Classifier Train/test split Label encoding Model evaluation (accuracy, confusion matrix)	 Inim...	 Ei ol... ▾	Mirelle: 3 hours Anne-Liine : 3 hours
5. Visualization, Conclusions, and Final Presentation Create final visuals, prepare the poster, summarize insights, and present results. - Avoid being too broad; focus on specific customer types and shopping behaviors. - Highlight how results help businesses make ethical, efficient decisions.	Matplotlib charts Table summaries	 Inim...	 Ei ol... ▾	Mirelle: 2 hours Anne-Liine : 2 hours